SYSTEM DOCUMENTATION · PUBLIC EDITION · ACCOUNTS PAYABLE AUTOMATION

Invoice & Expense Processing Automation

An n8n workflow that reads vendor invoices from email or a web form, extracts the details with AI, checks every number in code, asks the owner to approve anything over $500, files the PDF and records every outcome in a Google Sheets register.

V1 · built and tested · 1 Oct 2026 | n8n (self-hosted) | OpenRouter · gpt-4.1-mini | Gmail | Google Sheets | Google Drive

Demo business: Brightside Creative Studio (fictional). All vendors, invoices and amounts are test data. Built by Khael Mendoza · KhaelWorks. Figures marked "estimate" are modelled, not measured client results.

10 / 10	7–16 s	~$0.0004	$500
test invoices handled as expected on the final run	per invoice, receipt to Register (measured, excluding approval wait)	AI cost per invoice (estimate)	approval limit; every decision saved with approver and time

CLIENT QUESTIONS

What clients ask about this system

01 Who needs it?

Small and mid-sized businesses that receive roughly **20 to 500 vendor invoices a month** and handle them by hand: agencies, clinics, e-commerce shops, property managers, coaches and consultants with contractors, and the bookkeepers or virtual assistants who do accounts payable for them. It suits teams where one person keys invoices and an owner approves spending.

02 Why do companies need this?

Manual accounts payable leaks money and time in predictable ways:

- **Duplicate payments** when a vendor resends an invoice as a "reminder"
- **Wrong totals paid** because nobody re-adds the vendor's math
- **Spending without approval**, or approvals lost in chat with no record of who said yes
- **Hours of data entry** each week, and PDFs scattered across inboxes
- **Late payments** because invoices sit unread until the weekly catch-up

This system closes each of those gaps with a specific check, and leaves a record for every invoice.

03 How does it work?

1. **Intake.** An invoice arrives by email (subject contains "invoice" + PDF attached) or through an upload form.
2. **Read.** The PDF's text is extracted. Scanned images with no text go straight to human review.
3. **Extract.** AI turns the text into fields: vendor, invoice number, dates, subtotal, tax, total, category. It must return "not found" rather than guess.
4. **Check.** Code validates everything: required fields, subtotal + tax = total, sensible dates, currency, and that the AI's invoice number and total really appear in the PDF. Then it checks the Register for duplicates.
5. **Approve.** Totals over $500 trigger an email to the owner with Approve and Reject buttons. The workflow waits for the click.
6. **File and log.** The PDF is renamed and filed in Drive, and one row is written to the Register with the outcome, the approver, the time and a link to the PDF.

A full diagram is in "How it works", below.

04 How does it cut the invoice cycle from about a week to a day?

Most of the delay in manual accounts payable is **waiting**, not working: invoices wait in the inbox for a weekly catch-up, then wait again for the owner to approve. The automation removes the first wait completely and shortens the second to one click. The example below is **illustrative** (a business that does accounts payable in a weekly batch), not a measured client result.

STEP	MANUAL (WEEKLY BATCH)	WITH THE AUTOMATION
Invoice arrives	Monday, sits unread in the inbox	Monday, picked up within about a minute
Keyed in and checked	Friday batch, about 8 minutes per invoice	Done in seconds (7–16 s measured)
Duplicate check	From memory, if at all	Automatic against every live record
Approval request	Friday, by chat message	Monday, the same minute, by email with buttons
Approval received	Next week, when the owner replies	When the owner clicks, typically the same day
Filed and recorded	Whenever someone remembers	Immediately, with a link to the PDF
Arrival to approved and recorded	**Up to about 7 days**	**Same day; seconds for invoices of $500 or less**

Hands-on time drops from about 8 minutes per invoice to about 1 minute on average, because people only touch approvals and exceptions (estimate; see "Business impact").

05 Does the AI decide what gets paid?

No. The design rule is **AI reads, code checks, humans approve.** The AI only converts text into fields. Fixed code rules decide whether an invoice passes, and a person approves every invoice over the limit.

06 What if the AI reads something wrong?

The code catches it before it reaches the Register as approved. Every total is re-added; the AI's invoice number and total must appear word for word in the PDF text; missing fields fail validation. In testing, the AI returned "not found" for a missing invoice number instead of inventing one, and copied a wrong vendor total exactly as printed, which the math check then flagged.

07 Does it pay vendors automatically?

No, by design. It produces a checked, approved list with links to the PDFs. Payment stays in the company's bank or accounting tool, under the owner's control.

08 What happens when a vendor sends the same invoice twice?

It is marked **Duplicate**, with a note naming the original record, and filed for review instead of being approved again. The match is vendor + invoice number, ignoring spacing and punctuation, and it also catches two copies arriving in the same run.

09 Can it read scanned or photographed invoices?

Not in V1. They are detected (no readable text), sent to **Needs Review** with a clear note, and never sent to the AI, so nothing is guessed. Adding OCR is the first item on the roadmap.

10 What does a business need to run it?

A Google account (Gmail, Drive, Sheets), an n8n instance (self-hosted on a small VPS, or n8n Cloud) and an OpenRouter API key. No new software for the team: they keep using email, a spreadsheet and Drive.

11 How much does it cost to run?

The AI costs about **$0.0004 per invoice** (about 650 tokens on gpt-4.1-mini at OpenRouter list prices; estimate), so roughly 4 cents per 100 invoices. The other running costs are the n8n hosting the business already pays for or chooses, and its existing Google account.

12 Is our data safe? Where does it live?

Invoices, PDFs and the Register stay in the company's own Google account. n8n runs on a server the business controls. Only the invoice **text** is sent to the AI model for extraction. n8n keeps run logs on that same server for troubleshooting, and they can be set to expire. Access uses Google sign-in (OAuth), not shared passwords.

13 Can we change the approval limit, approvers or categories?

Yes. The limit, approver email, currency and the category list are single settings in the workflow. Each change is tested with one invoice before going live.

14 Does it connect to QuickBooks or Xero?

Not in V1; the Register is the hand-off point. Creating bills in QuickBooks or Xero for approved invoices is on the roadmap.

15 What happens if something breaks?

The AI, Google Sheets, Google Drive and Gmail-label steps each retry 3 times. If a run still fails, a separate alert workflow emails the owner the failed step, the error, a plain-English likely fix and a link to the run. Nothing is lost: the original email or form submission remains and the run can be retried.

16 How many invoices can it handle?

Email is checked every minute, one invoice email per check, so up to about 60 emailed invoices an hour; form uploads run independently. That is far above the volume of the businesses it targets. Larger volumes would move approvals into a sub-workflow (roadmap).

17 What does our team still do?

Approve or reject invoices over the limit (one click each), clear the weekly Needs Review list, and pay vendors as usual. An Owner SOP covers each task step by step.

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18 How long does setup take for a new business?

This V1 was built and tested in 7 days. Adapting it to a new business (its inbox, approvers, categories, currency and a test run with its own sample invoices) is estimated at about **1 to 2 weeks**, depending on how many approvers and categories it needs.

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19 Can it handle other currencies, such as PHP?

Yes, one currency per setup in V1 (set in a single place). Invoices in any other currency are flagged for review rather than processed.

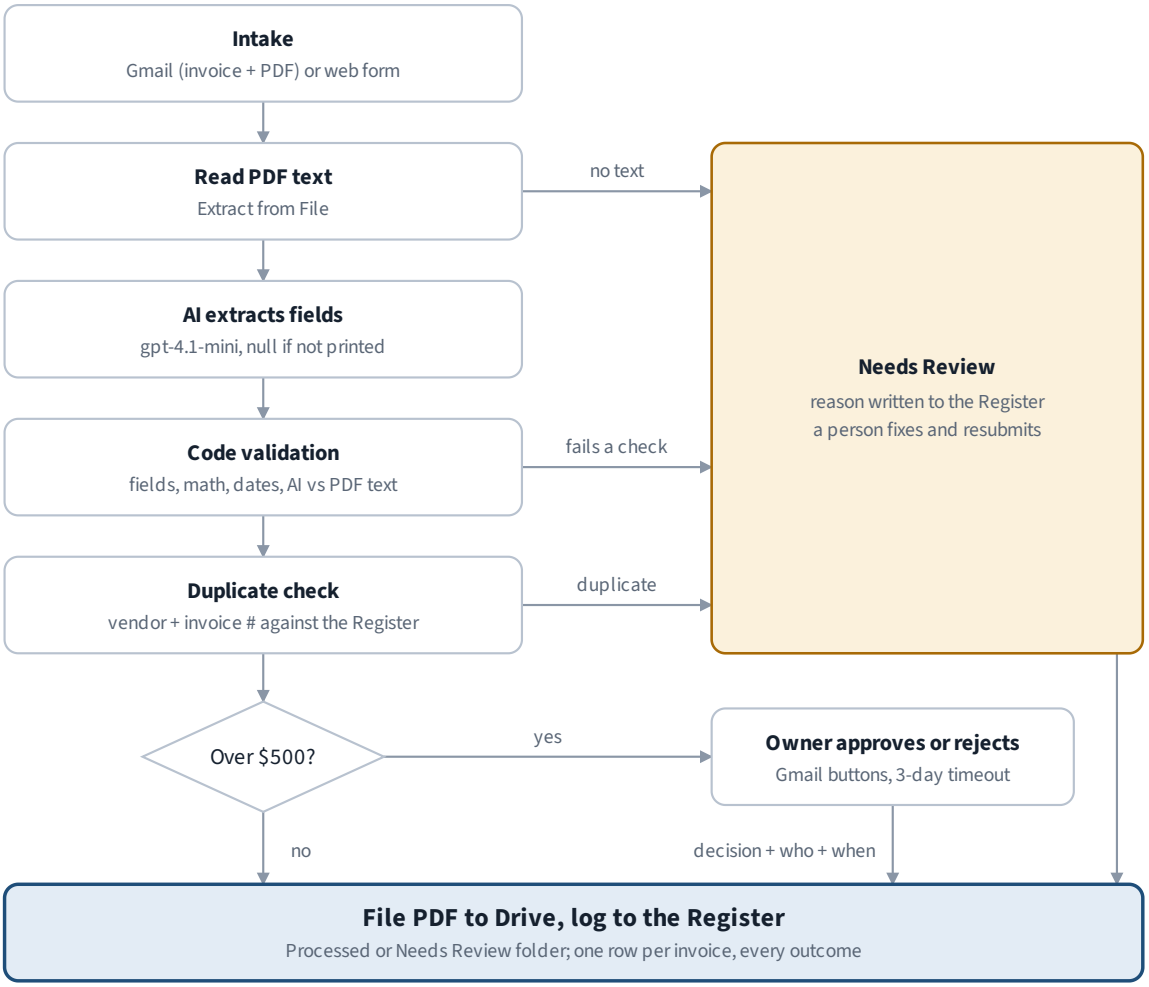
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20 How was it tested?

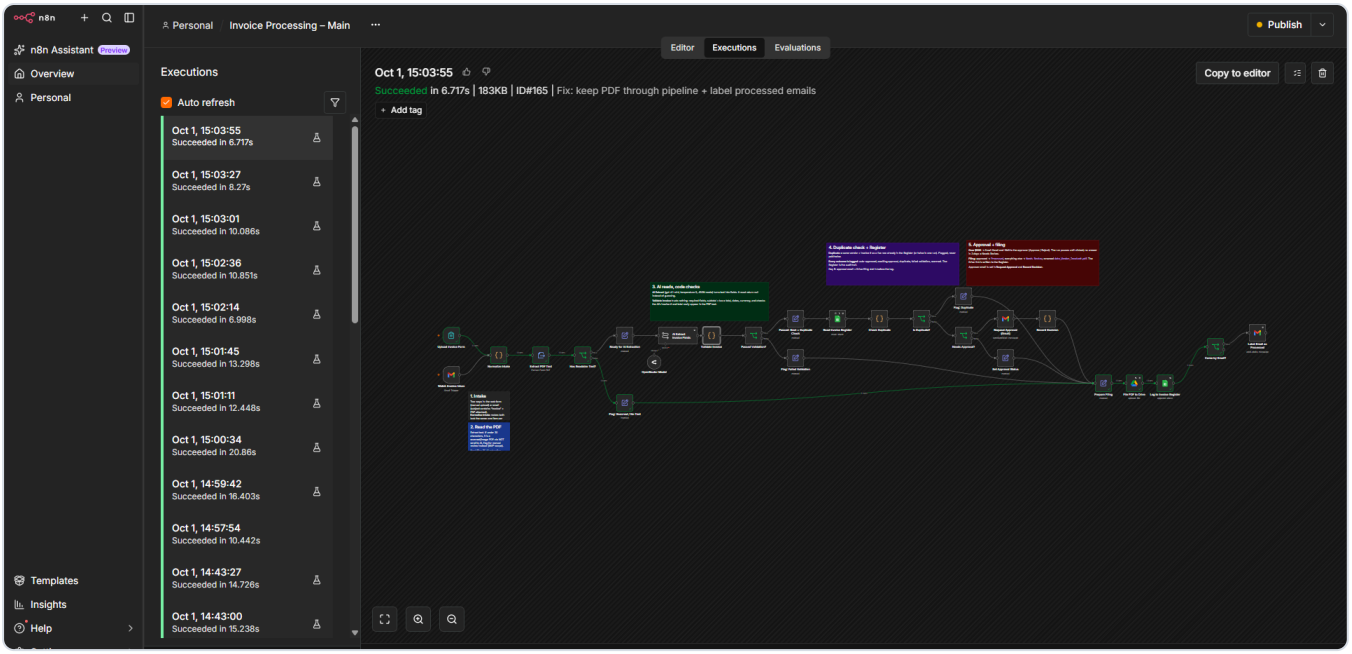
With 10 purpose-built test invoices, each designed to exercise one path: clean, over the limit (approved and rejected), exactly at the limit, a math error, a missing invoice number, a future date, a resent duplicate, a scanned image and an emailed invoice. All 10 matched the expected outcome on the final live run. Details are in "Test results".

HOW IT WORKS

Every invoice ends in the Register; only clean ones skip a human



Two workflows: **Invoice Processing – Main** (31 nodes, including notes) and **Invoice Processing – Error Alerts** (runs only when Main fails).



The full main workflow in n8n, with the final live test run selected (every execution succeeded). Intake on the left, AI and checks in the middle, approval, filing and logging on the right.

CONTROLS

AI reads, code checks, humans approve

CHECK	RULE	CAUGHT IN TESTING
Readable PDF	Fewer than 30 characters of text → manual review, AI skipped	#10 scanned image
AI output	Temperature 0, JSON only, null when not printed, fixed category list	#6 returned null, not an invented number
Required fields	Vendor, invoice #, invoice date, total	#6 missing invoice number
Math	Subtotal + tax = total, within $0.01	#5 total $20.00 too high
Dates	Valid; not in the future; not over a year old; due date not before invoice date	#8 dated 2027-01-15
Currency	Must match the configured currency (USD)	—
Hallucination guard	AI's invoice # and total must appear in the PDF text	—
Duplicate	Vendor + invoice # against live Register rows and within the same run	#7 resend of #1
Approval	Total over $500 → Approve / Reject email; 3-day timeout → review	#2 approved, #4 rejected, #9 at $500 not sent

RELIABILITY

No invoice disappears silently

FAILURE	WHAT HAPPENS
AI, Sheets, Drive or Gmail-label step fails	Retries 3 times, 2 seconds apart; then an alert email
AI returns unusable data	Logged as Needs Review; nothing guessed
Drive upload fails	Invoice still logged, with "Drive upload failed – PDF not filed"
Approver never responds	After 3 days the invoice moves to Needs Review
Same email seen twice	Gmail label added after logging; labelled mail is skipped
Any unhandled error	Error Alerts workflow emails the failed step, error, likely fix and run link

TESTING

Test results: 10 of 10 as expected

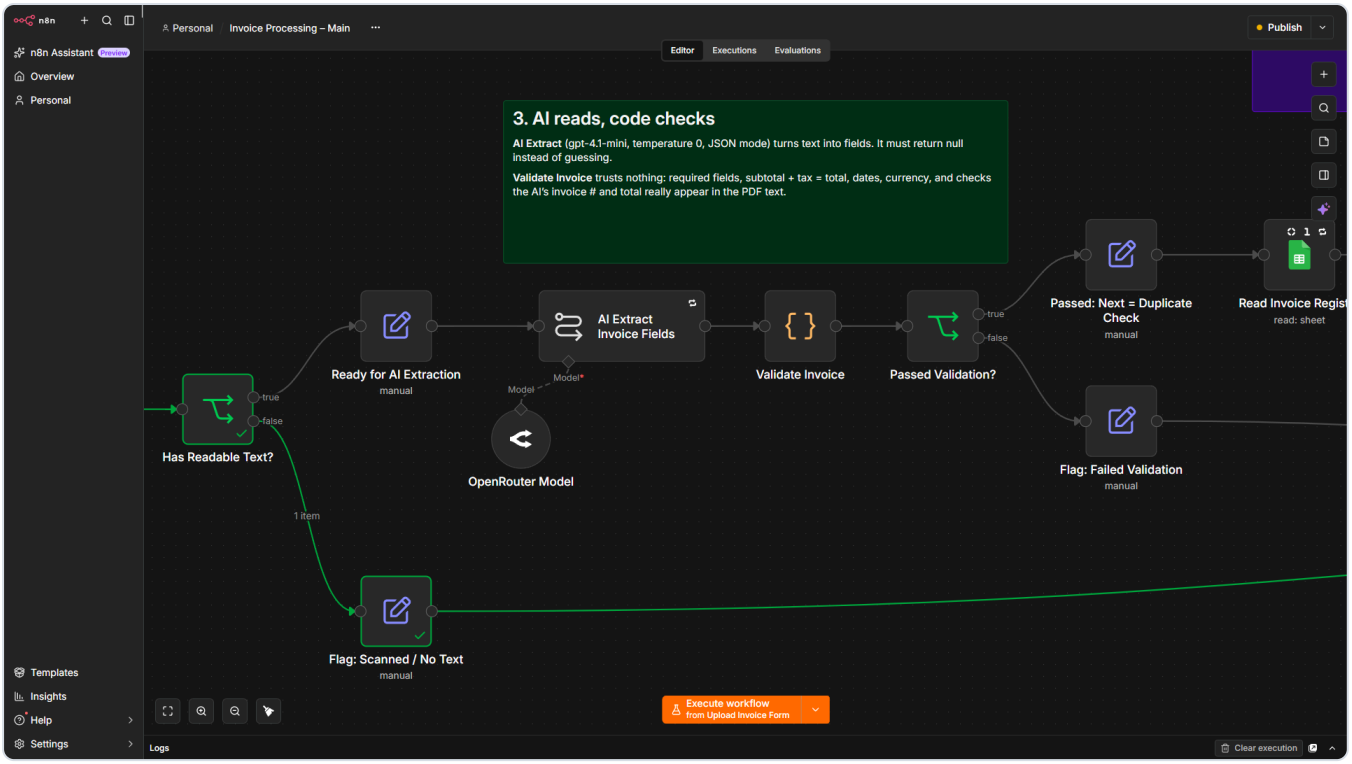
#	VENDOR (FICTIONAL)	TOTAL	TESTS	RESULT	
1	Summit Office Supply Co.	$189.87	Clean, under limit	Approved (auto), Office Supplies	Pass
2	CloudDesk Software LLC	$1,286.01	Over limit	Approved by owner via email	Pass
3	Metro Fiber Internet	$113.65	Email intake	Approved (auto); email labelled, not reprocessed	Pass
4	Rivera Web Design	$2,550.00	Over limit, $0 tax	Rejected by owner; filed as REJECTED	Pass
5	Greenleaf Catering	$474.65	Math error	Needs Review: 420.00 + 34.65 = 454.65	Pass
6	Apex Print & Sign	$308.51	Missing invoice #	Needs Review; AI did not invent one	Pass
7	Summit Office Supply Co.	$189.87	Resent duplicate	Duplicate of #1's record	Pass
8	Harbor Courier Services	$146.14	Future date	Needs Review: dated 2027-01-15	Pass
9	Pixel & Pine Stock Media	$500.00	Exactly at limit	Approved (auto): $500 is not over $500	Pass
10	Lakeside Janitorial	$346.40	Scanned image	Needs Review: no readable text	Pass

Bugs found and fixed during the build, before the final run: AI output read from the wrong field (all invoices failed validation), the PDF file dropped mid-pipeline (nothing filed to Drive), a Google API left disabled, and identical record IDs within one batch. Each was caught by a real test run.

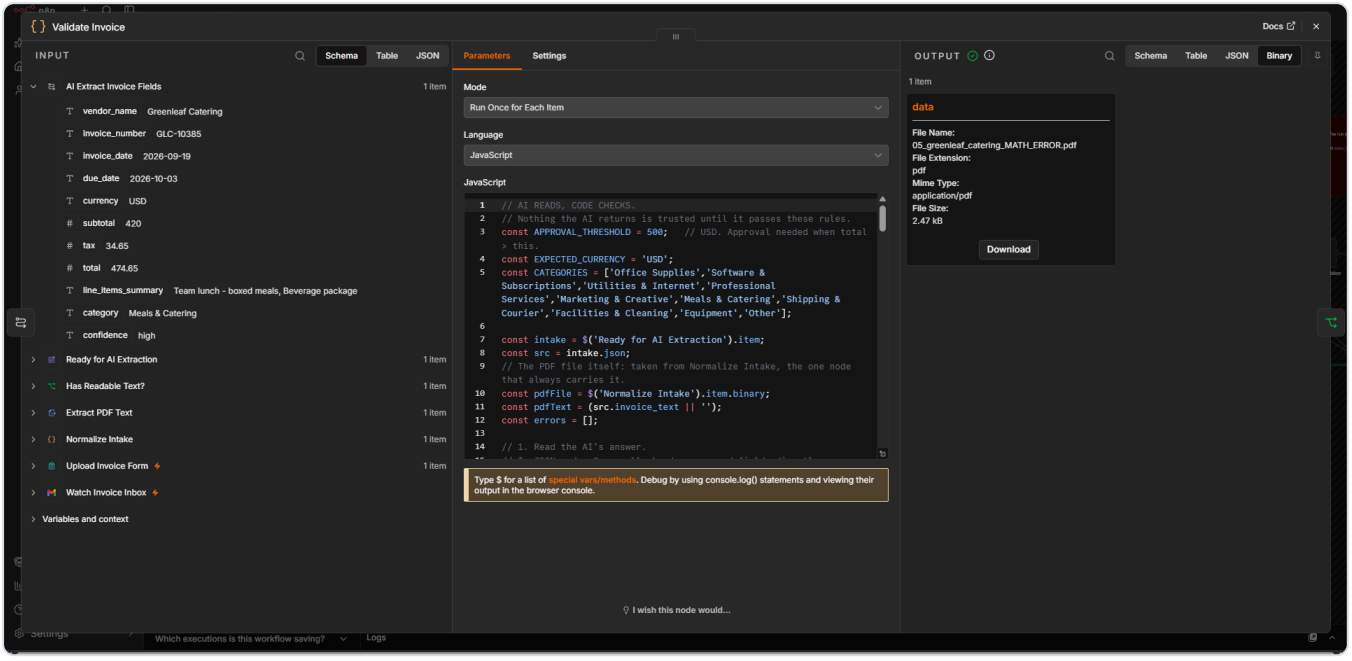
EVIDENCE

Screens from the test runs

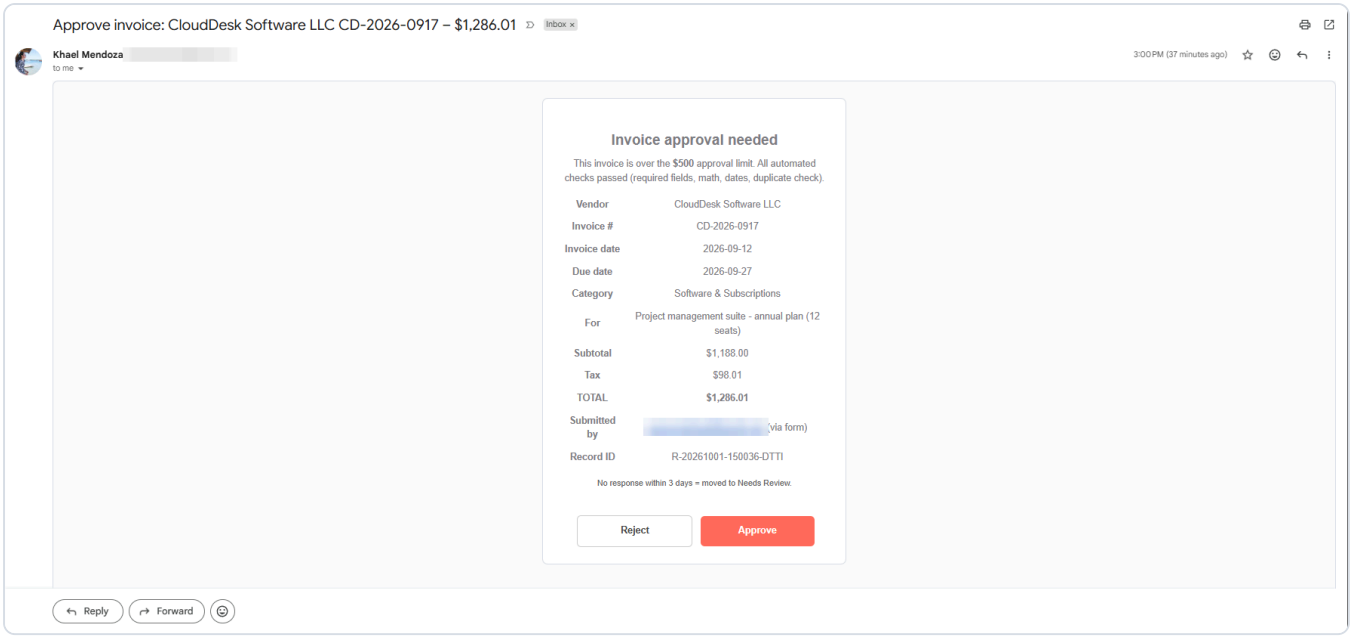
Screenshots from the live system. Email addresses and the server address are blurred or cropped.



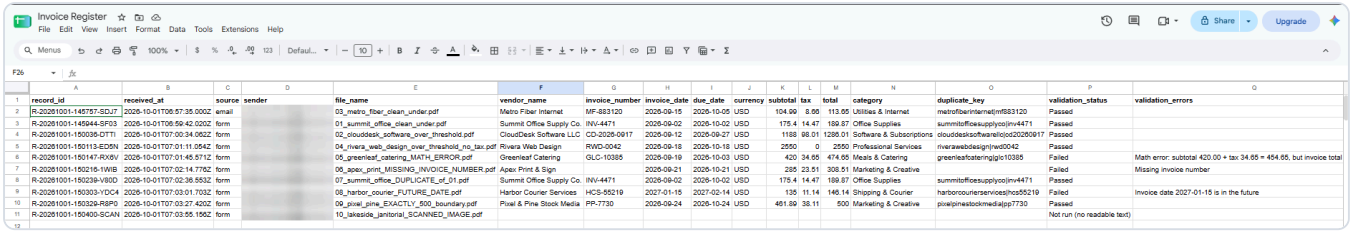
The AI and validation stage. Scanned PDFs branch off before the AI; anything failing a check branches to review.



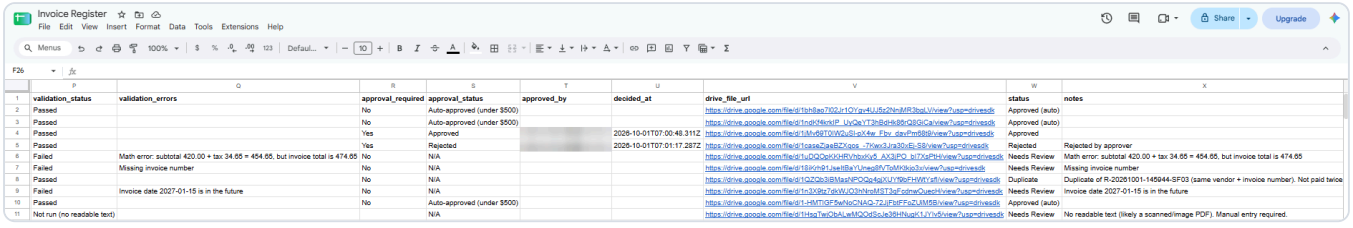
Inside Validate Invoice: the AI's fields arrive on the left (here the math-error invoice, #5); the rules are plain code; the original PDF is carried through for filing.



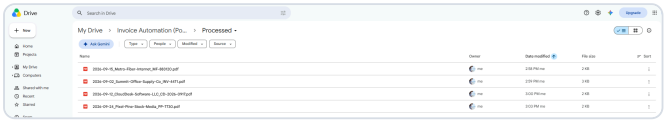
Approval email for an invoice over $500. One click records the decision, the approver and the time.



Invoice Register, left half: one row per invoice from the final run, with the extracted fields and category.



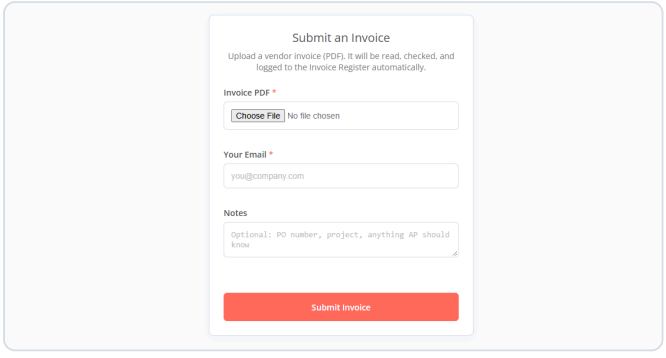
Invoice Register, right half: approval, approver, decision time, Drive link, status and the reason for every exception.



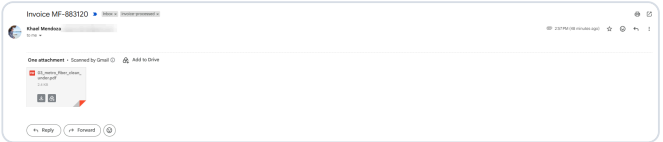
Processed folder: approved PDFs, renamed date_Vendor_Invoice#.



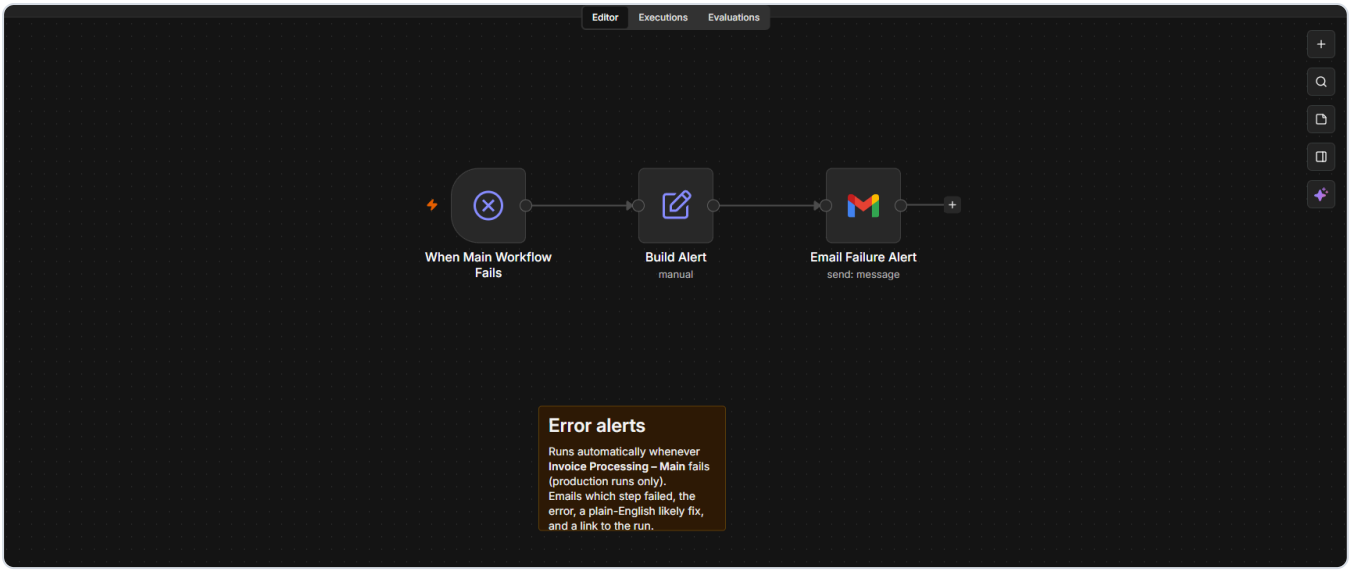
Needs Review folder: rejected, failed, duplicate and scanned invoices, each marked in the file name.



The upload form staff use.



An emailed invoice after processing: labelled invoice-processed so it is never picked up twice.



The Error Alerts workflow, which runs only when the main workflow fails.

BUSINESS IMPACT · ESTIMATES

About 12 staff hours a month saved at 100 invoices

Assumptions: about 8 minutes of manual handling per invoice; about 20% need approval (about 1 minute each); about 15% need review (about 5 minutes each); AI at about $0.0004 per invoice. These are estimates, not measured client results.

INVOICES / MONTH	MANUAL TIME	WITH AUTOMATION	TIME SAVED (EST.)	AI COST (EST.)
50	6.7 h	0.8 h	5.9 h	$0.02
100	13.3 h	1.6 h	11.8 h	$0.04
300	40.0 h	4.8 h	35.3 h	$0.12

Not priced but reduced: duplicate payments, overpaid vendor errors, spending without a recorded approval, and time spent hunting for PDFs. To measure a real client's numbers: time 10 manual invoices before go-live, then compare with the first month of n8n executions.

SCOPE

What V1 does not do yet

LIMIT TODAY	PLANNED
Scanned or photographed invoices go to manual entry	OCR step before the AI
Google sign-in renews weekly (app in testing mode)	Publish the Google app with a privacy policy
One invoice email per check	Approvals in a sub-workflow for batches
Settings live inside the workflow	A Config tab in the Register
No accounting-system sync	QuickBooks / Xero bill creation for approved invoices

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Built with n8n, OpenRouter, Gmail, Google Sheets and Google Drive